

Tools and Environment

To Verify environment:

```
source ./verify_env.sh
```

Example Output

```
=== cocotb_modified3 Environment Verification ===  
[Python]  
Python 3.11.8  
cocotb present: True  
[Verilator]  
Verilator 5.040 2025-08-30 rev v5.040  
[GTKWave]  
GTKWave Analyzer v3.3.116 (w)1999-2023 BSI
```

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```
[Build Tools]  
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0  
g++ (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0  
GNU Make 4.3  
1.8.1  
=== Verification Complete ===
```

Quick check Sheet

Task	Command
Recreate venv	<code>python3.11 -m venv py311</code>
Activate venv	<code>source py311/bin/activate</code>
List venv packages	<code>pip list</code>
List system Python 3.11 packages	<code>python3.11 -m pip list</code>

Recreate & Activate

From your project root (cocotb_verif_accumulator):

bash

```
rm -rf py311                (delete old files)
python3.11 -m venv py311    (create environment)
source py311/bin/activate   (activate environment)
```

Always activate environment "source py311/bin/activate" before executing tests in the verification project or configuring - should be the first thing done in any use case.

Install Required Packages:

bash

```
pip install --upgrade pip
pip install cocotb cocotb-tools verilator pytest
pip install -r requirements.txt
```

Note "requirements.txt" is a document in the project root directory. There is no need to install the items 1 by 1 the previous command install the following packages :

```
cocotb==2.0.1
cocotb-test==0.2.2
cocotb-coverage==1.1.0

pytest==7.4.4
pytest-xdist==3.5.0

find-libpython==0.4.0
setuptools>=68
wheel>=0.42

# Optional but recommended
pyuv==2.8.0
numpy==1.26.4
rich==13.7.1
packaging==24.0
```

2. List installed Python packages inside py311

Activate the environment:

```
source py311/bin/activate
```

List packages:

```
pip list
```

Or List with versions and locations:

```
pip list --verbose
```

Freeze for reproducibility:

```
pip freeze
```

Shows **system-wide** Python 3.11 packages (not just the venv):

```
python3.11 -m pip list  
or python3.11 -m pip freeze
```

If you want to see where they live:

```
python3.11 -m pip list --verbose
```

This will show:

- system site-packages
- user site-packages
- anything installed outside your venv
-

List all Python 3.11 standard library modules (recommended)

- Run this inside your py311 environment:
- bash
- python3.11 - << 'EOF'
- import sysconfig, pkgutil
-
- stdlib = sysconfig.get_paths()["stdlib"]
- modules = sorted(name for _, name, _ in pkgutil.iter_modules([stdlib]))
-
- for m in modules:
- print(m)
- EOF

List built-in C modules (compiled into Python)

Run:

```
bash
python3.11 - << 'EOF'
import sys
for m in sys.builtin_module_names:
    print(m)
EOF
```

Examples from Pc that developed this project:

cocotb_verif_accumulator \$ pip list

Package	Version
cocotb	2.0.1
cocotb-coverage	1.1.0
contourpy	1.3.3
cycler	0.12.1
find_libpython	0.5.1
fonttools	4.63.0
iniconfig	2.3.0
kiwisolver	1.5.0
matplotlib	3.11.0
numpy	2.4.6
packaging	26.2
pillow	12.2.0
pip	26.1.2
pluggy	1.6.0
Pygments	2.20.0
pyparsing	3.3.2
pytest	9.1.1
python-constraint	1.4.0
python-dateutil	2.9.0.post0
PyYAML	6.0.3
setuptools	65.5.0
six	1.17.0